

# Progress Report

## Overview

We have successfully implemented **Trust Boundary 3**, which represents the secure internal zone of the cloud architecture. This layer includes the **Application Server** and four **Distributed Storage Servers**. The focus of this phase was secure design, authentication, authorization and logging.

## Application Server Implementation

The Application Server manages file metadata, chunk allocation, and coordination with storage servers.

### Security features implemented:

- Token-based authentication system
- Role-Based Access Control (Admin, User, ReadOnly)
- Rate limiting (60 requests per minute per IP)
- Account lockout after 5 failed login attempts (5-minute lock)
- Token expiry enforcement
- IP-bound session validation
- Separate authentication, threat, and operation logs

## Storage Servers Implementation

Four storage servers were implemented to support distributed file storage and retrieval.

### Security features implemented:

- SHA-256 integrity verification for all stored chunks
- Maximum chunk size enforcement (10 MB)
- Rate limiting (100 requests per minute per IP)
- Detailed audit logging of all operations
- Persistent checksum storage for integrity validation

## Current Status

- Application Server fully implemented and secured
- Four Storage Servers implemented with integrity protection
- Comprehensive logging across all components
- System runnable across configured machines

Trust Boundary 3 is complete, secure, and ready for integration with the remaining architecture layers.